VLSI Design Lab (ECN15106) · Cheat sheet · Verilog

Exp 1 · 4-bit Adder-Subtractor

M=0 add, M=1 sub: A−B = A+$\bar{B}$+1. Cout=1 in sub ⇒ A≥B. Out: 8,15,1,2,4,4,0,12.

DESIGN

```
`timescale 1ns / 1ps
module Add_Sub_G5(
    input  [3:0] A,
    input  [3:0] B,
    input        M,
    output reg [3:0] S,
    output reg   Cout
);
reg [4:0] temp;
reg [3:0] bcomp;
always @(*) begin
    bcomp = B ^ {4{M}};          // invert B when M = 1
    temp  = A + bcomp + M;       // M is also the carry-
in (+1)
    Cout  = temp[4];             // 5th bit = carry out
    if (M && (B > A))
        S = ~(temp[3:0]) + 4'b0001;   // negative answer
-> show magnitude
    else
        S = temp[3:0];
end
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module tb_Add_Sub_G5;
reg  [3:0] A;
reg  [3:0] B;
reg        M;
wire [3:0] S;
wire       Cout;
Add_Sub_G5 uut(.A(A), .B(B), .M(M), .S(S), .Cout(Cout));
initial begin
    $display("Time\tA\tB\tM\tS\tCout");
    $monitor("%0t\t%d\t%d\t%b\t%d\t%b", $time, A, B, M,
S, Cout);
    A = 4'd5;  B = 4'd3; M = 0; #10;
    A = 4'd9;  B = 4'd6; M = 0; #10;
    A = 4'd12; B = 4'd5; M = 0; #10;
    A = 4'd6;  B = 4'd4; M = 1; #10;
    A = 4'd3;  B = 4'd7; M = 1; #10;
    A = 4'd10; B = 4'd6; M = 1; #10;
    A = 4'd9;  B = 4'd9; M = 1; #10;
    A = 4'd7;  B = 4'd5; M = 0; #10;
    $finish;
end
endmodule
```

Exp 2 · Degree → Radian

K=(π/180)·2^16≈1144; rad = deg·1144/65536. 30°→0.524, 90°→1.571, 180°→3.142, 360°→6.284.

DESIGN

```
`timescale 1ns / 1ps
module deg_rad_G5(
    input  [15:0] degree,
    output [31:0] radian
);
parameter K = 16'd1144;          // (pi/180) x 2^16
assign radian = degree * K;      // fixed-point result,
Q16
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module tb_deg_rad_G5;
reg  [15:0] degree;
wire [31:0] radian;
real radian_real;
deg_rad_G5 uut(.degree(degree), .radian(radian));
always @(*) begin
    radian_real = $rtoi((radian / 65536.0) * 1000.0 +
0.5) / 1000.0;
end
initial begin
    $display("Time\tDegree\tRadian");
    $monitor("%0t\t%d\t%f", $time, degree, radian_real);
    degree = 0;   #10;
    degree = 30;  #10;
    degree = 45;  #10;
    degree = 60;  #10;
    degree = 90;  #10;
    degree = 120; #10;
    degree = 180; #10;
    degree = 360; #10;
    $finish;
end
endmodule
```

Exp 3 · MOD-13 Counter (T-FF)

T0=(Q3Q2)', T1=Q0, T2=Q3Q2+Q1Q0, T3=Q3Q2+Q2Q1Q0. 0→12→0. Unused 13→3, 14→2, 15→1.

DESIGN

```
`timescale 1ns / 1ps
module mod13_up_counter_G5(
    input  clk,
    input  reset,
    output [3:0] Q
);
    // record names T1..T4 for flip-flops Q0..Q3
    wire T1 = ~Q[3] | ~Q[2];                          //
T0 = (Q3Q2)'
    wire T2 = Q[0];                                   //
T1 = Q0
    wire T3 = (Q[3] & Q[2]) | (Q[1] & Q[0]);          //
T2 = Q3Q2 + Q1Q0
    wire T4 = (Q[3] & Q[2]) | (Q[2] & Q[1] & Q[0]);   //
T3 = Q3Q2 + Q2Q1Q0
    T_FF_G5 t1 (clk, reset, T1, Q[0]);
    T_FF_G5 t2 (clk, reset, T2, Q[1]);
    T_FF_G5 t3 (clk, reset, T3, Q[2]);
    T_FF_G5 t4 (clk, reset, T4, Q[3]);
endmodule
`timescale 1ns / 1ps
module T_FF_G5(
    input  clk,
    input  reset,
    input  T,
    output reg Q
);
always @(posedge clk or posedge reset) begin
    if (reset)
        Q <= 1'b0;        // asynchronous reset
    else if (T)
        Q <= ~Q;          // toggle
    else
        Q <= Q;           // hold
end
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module tb_mod13_up_counter_G5;
reg  clk;
reg  reset;
wire [3:0] Q;
mod13_up_counter_G5 DUT(.clk(clk), .reset(reset), .Q(Q));
always #5 clk = ~clk;            // 10 ns period = 100
MHz
initial begin
    clk   = 1'b0;
    reset = 1'b1;
    #10;
    reset = 1'b0;
    #150;
    $finish;
end
initial begin
    $monitor("Time = %0t | Reset = %b | Q = %d (%b)",
$time, reset, Q, Q);
end
endmodule
```

Exp 4 · Full Adder on FPGA

S=A⊕B⊕Cin, Cout=AB+BCin+ACin. Pins: a V17(SW0), b V16(SW1), cin W16(SW2) → sum U16(LD0), cout E19(LD1).

DESIGN

```
`timescale 1ns / 1ps
module full_adder(
    input  a,        // SW0
    input  b,        // SW1
    input  cin,      // SW2
    output sum,      // LD0
    output cout      // LD1
);
assign sum  = a ^ b ^ cin;
assign cout = (a & b) | (b & cin) | (a & cin);
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module tb_full_adder;
reg  a, b, cin;
wire sum, cout;
integer i;
full_adder uut(.a(a), .b(b), .cin(cin), .sum(sum),
.cout(cout));
initial begin
    $display("a b cin | sum cout");
    for (i = 0; i < 8; i = i + 1) begin
        {a, b, cin} = i[2:0];
        #10;
        $display("%b %b  %b  |  %b    %b", a, b, cin,
sum, cout);
    end
    $finish;
end
endmodule
```

Exp 6 · Binary → Gray on FPGA

G3=B3, Gi=Bi+1⊕Bi. Pins: a[0..3] V17,V16,W16,W17 → gray[0..3] U16,E19,U19,V19.

DESIGN

```
`timescale 1ns / 1ps
module binary_to_gray (gray, a);
output [3:0] gray;
input  [3:0] a;
assign gray[3] = a[3];
assign gray[2] = a[3] ^ a[2];
assign gray[1] = a[2] ^ a[1];
assign gray[0] = a[1] ^ a[0];
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module testbench;
reg  [3:0] A;
wire [3:0] GRAY;
binary_to_gray BG (GRAY, A);     // connected by
position: (gray, a)
initial begin
    $monitor($time, " BINARY= %b, GRAY= %b", A, GRAY);
end
initial begin
    A = 4'd0;
    #5 A = 4'd1;
    #5 A = 4'd2;
    #5 A = 4'd3;
    #5 A = 4'd4;
    #5 A = 4'd5;
    #5 A = 4'd6;
    #5 A = 4'd7;
    #5 A = 4'd8;
    #5 A = 4'd9;
    #5 A = 4'd10;
    #5 A = 4'd11;
    #5 A = 4'd12;
    #5 A = 4'd13;
    #5 A = 4'd14;
    #5 A = 4'd15;
    #5 $finish;
end
endmodule
```

Exp 8 · 4-bit Adder on FPGA

4 FA ripple carry. a[0..3] V17,V16,W16,W17; b[0..3] V2,T3,T2,R3; c_in R2 → sum[0..3] U16,E19,U19,V19; c_out U15(LD5).

DESIGN

```
`timescale 1ns / 1ps
module fulladd (sum, c_out, a, b, c_in);
output sum, c_out;
input  a, b, c_in;
wire s1, c1, c2;
xor (s1, a, b);          // half adder 1: sum
and (c1, a, b);          // half adder 1: carry
xor (sum, s1, c_in);     // half adder 2: sum
and (c2, s1, c_in);      // half adder 2: carry
or  (c_out, c2, c1);     // final carry
endmodule
module fulladd4 (sum, c_out, a, b, c_in);
output [3:0] sum;
output c_out;
input  [3:0] a, b;
input  c_in;
wire c1, c2, c3;
fulladd fa0(sum[0], c1,    a[0], b[0], c_in);
fulladd fa1(sum[1], c2,    a[1], b[1], c1);
fulladd fa2(sum[2], c3,    a[2], b[2], c2);
fulladd fa3(sum[3], c_out, a[3], b[3], c3);
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module testbench;
reg  [3:0] A, B;
reg  C_IN;
wire [3:0] SUM;
wire C_OUT;
fulladd4 FA_4 (SUM, C_OUT, A, B, C_IN);
initial begin
    $monitor($time, " A= %b, B= %b, C_IN= %b, C_OUT= %b,
SUM= %b",
             A, B, C_IN, C_OUT, SUM);
end
initial begin
    A = 4'd0;  B = 4'd0; C_IN = 1'b0;
    #5 A = 4'd3;  B = 4'd4;
    #5 A = 4'd2;  B = 4'd3;
    #5 A = 4'd2;  B = 4'd2;
    #5 A = 4'd0;  B = 4'd3;
    #5 A = 4'd10; B = 4'd5; C_IN = 1'b1;
    #5 $finish;
end
endmodule
```

Exp 10 · 4-bit Comparator

Ei=XNOR(Ai,Bi). Eq=E3E2E1E0. Gt=A3$\bar{B}$3+E3A2$\bar{B}$2+E3E2A1$\bar{B}$1+E3E2E1A0$\bar{B}$0. MSB first.

DESIGN

```
`timescale 1ns / 1ps
module four_bit_digital_comparator_G5(
    input  [3:0] A,
    input  [3:0] B,
    output A_greater_B,
    output A_lesser_B,
    output A_equal_B
);
    wire E3 = ~(A[3] ^ B[3]);    // XNOR: 1 when bits are
equal
    wire E2 = ~(A[2] ^ B[2]);
    wire E1 = ~(A[1] ^ B[1]);
    wire E0 = ~(A[0] ^ B[0]);
    assign A_greater_B = (A[3] & ~B[3]) |
                         (E3 & A[2] & ~B[2]) |
                         (E3 & E2 & A[1] & ~B[1]) |
                         (E3 & E2 & E1 & A[0] & ~B[0]);
    assign A_lesser_B  = (~A[3] & B[3]) |
                         (E3 & ~A[2] & B[2]) |
                         (E3 & E2 & ~A[1] & B[1]) |
                         (E3 & E2 & E1 & ~A[0] & B[0]);
    assign A_equal_B   = E3 & E2 & E1 & E0;
endmodule
```

TESTBENCH

```
`timescale 1ns / 1ps
module tb_four_bit_digital_comparator_G5;
reg  [3:0] A;
reg  [3:0] B;
wire A_greater_B, A_lesser_B, A_equal_B;
four_bit_digital_comparator_G5 uut (
    .A(A), .B(B),
    .A_greater_B(A_greater_B),
    .A_lesser_B(A_lesser_B),
    .A_equal_B(A_equal_B)
);
initial begin
    A = 4'b0000; B = 4'b0000; #10;
    A = 4'b1010; B = 4'b0101; #10;
    A = 4'b0101; B = 4'b1010; #10;
    A = 4'b1001; B = 4'b1001; #10;
    A = 4'b1000; B = 4'b0111; #10;
    A = 4'b0110; B = 4'b0101; #10;
    A = 4'b0100; B = 4'b0101; #10;
    A = 4'b0001; B = 4'b0000; #10;
    $finish;
end
initial begin
    $monitor("Time=%0t | A=%b B=%b | Greater=%b Lesser=%b
Equal=%b",
             $time, A, B, A_greater_B, A_lesser_B,
A_equal_B);
end
endmodule
```

XDC line format (Basys 3)

```
set_property -dict { PACKAGE_PIN V17 IOSTANDARD LVCMOS33
} [get_ports {a[0]}]
```

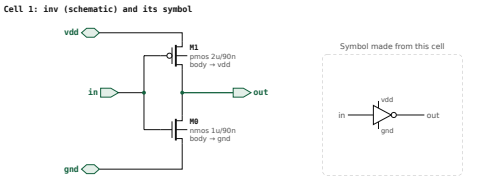
Flow: simulate → XDC → synthesis → implementation → bitstream → Hardware Manager → Program. Part xc7a35tcpg236-1.

Symica DE experiments (90 nm, VDD 1.2 V)

All Symica: .include 90nm_bulk.pm · nmos4/pmos4, model nmos/pmos · L 90n · bodies PMOS→vdd, NMOS→gnd · VDD 1.2 V · delay at 0.6 V · tr/tf 0.12→1.08 V · testbench = symbol + vdc + vpulse + CL + gnd.

Exp 7 · CMOS Inverter (DC + Tran)

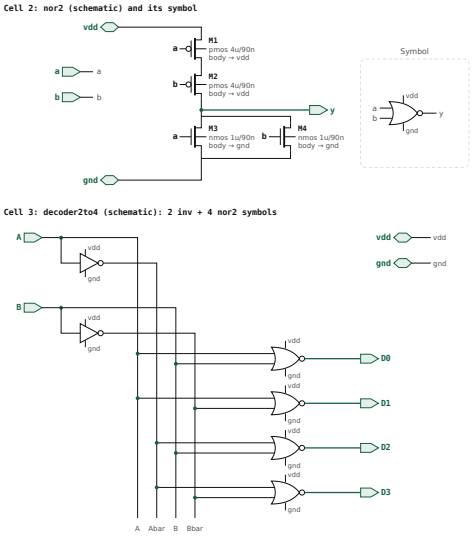
Regions NMOS: OFF→SAT→SAT→LIN→LIN (PMOS reverse). VM=VDD/2 if Wp≈2–3Wn. tp≈0.69RC, P=αCV²f.

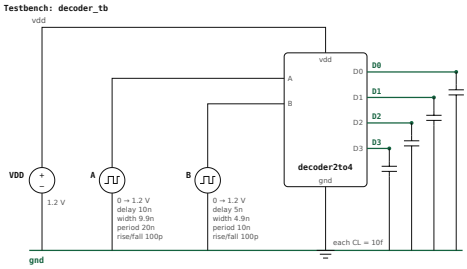


M1 pmos4	pmos 2u/90n, B→vdd
M0 nmos4	nmos 1u/90n, B→gnd
V0	vdc 1.2
V1	0→1.2, td 1n, pw 4.9n, per 10n, tr/tf 100p
CL	10f
DC	sweep V1 0→1.2, step 1m: VM, VIL/VIH (slope −1), NMH=VOH−VIH, NML=VIL−VOL
Tran	20n/10p: tpHL, tpLH, tr, tf

Exp 5 · CMOS 2-4 Decoder

D0=$\bar{A}\bar{B}$, D1=$\bar{A}$B, D2=A$\bar{B}$, D3=AB. 20 transistors. Rising output slower (series PMOS).

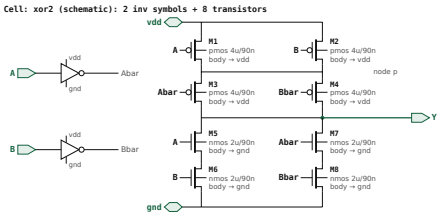




inv	pmos 2u / nmos 1u
nor2	PMOS series 4u, NMOS parallel 1u, L 90n
Wiring	D0=NOR(A,B) D1=NOR(A,$\bar{B}$) D2=NOR($\bar{A}$,B) D3=NOR($\bar{A}$,$\bar{B}$)
A	0→1.2, td 10n, pw 9.9n, per 20n, tr/tf 100p
B	0→1.2, td 5n, pw 4.9n, per 10n, tr/tf 100p
CL	10f ×4
Tran	40n/10p; 5ns B↑→D1↑,D0↓; 15ns B↑→D3↑,D2↓; glitch 10/20/30ns

Exp 9 · CMOS XOR + Delay

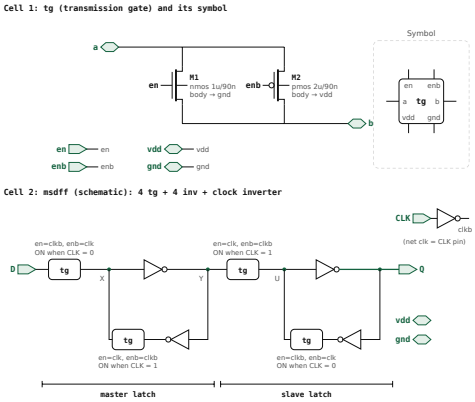
Y=$\bar{A}$B+A$\bar{B}$. PDN (A·B)‖($\bar{A}$·$\bar{B}$); PUN (A‖B) series ($\bar{A}$‖$\bar{B}$). 12T.



Core PMOS	4u/90n (M1–M4)
Core NMOS	2u/90n (M5–M8)
2×inv	2u/1u
A	0→1.2, td 2n, pw 4.9n, per 10n, tr/tf 100p
B	0→1.2, td 7n, pw 9.9n, per 20n, tr/tf 100p
CL	10f on Y
Tran	40n/10p; delay A↑/A↓ with B=0 and B=1 at 0.6 V

Exp 11 · Master-Slave D Flip-Flop

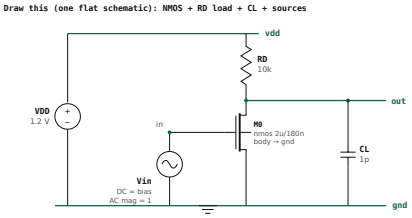
Master open CLK=0, slave CLK=1 → +edge. Tclk ≥ tc-q+tlogic+tsu. 18T.



tg	nmos 1u (en) ‖ pmos 2u (enb)
Master	tg(en=clkb) D→X, inv X→Y, inv Y→tg(en=clk)→X
Slave	tg(en=clk) Y→U, inv U→Q, inv Q→tg(en=clkb)→U
CLK	0→1.2, td 5n, pw 4.9n, per 10n, tr/tf 100p
D	0→1.2, td 2n, pw 7.9n, per 16n, tr/tf 100p
CL	10f on Q
Tran	60n/10p; tc-q, setup (move D toward edge), hold

Exp 12 · CS Amplifier (AC)

Av=−gm(RD‖ro), 180°. gm=2ID/Vov. f3dB=1/(2πRoutCL). Miller Cgd(1+|Av|).



M0	nmos 2u/180n, B→gnd
RD	10k vdd→out
CL	1p
VDD	1.2
Vin	DC bias (out≈0.6 V from DC sweep) + AC 1
OP	check VDS ≥ VGS−Vt
AC	1 Hz–10 GHz, dec 50: gain dB, f3dB, unity f, GBW